Question	Answer	Marks
4(a)	$\omega = 2\pi / T$	C1
	$= 2\pi / (0.15 \times 10^{-6}) = 4.2 \times 10^{7}\ \text{rad s}^{-1}$	A1
4(b)	$a_0 = \omega^2 x_0$	C1
	$= (4.2 \times 10^{7})^2 \times 40 \times 10^{-6}$ $= 7.1 \times 10^{10}\ \text{m s}^{-2}$	A1
4(c)	$E = ½ m\omega^2 x_0^2$	C1
	$= ½ \times 2.4 \times 10^{-4} \times (4.2 \times 10^{7})^2 \times (40 \times 10^{-6})^2$	C1
	$= 340\ \text{J}$	A1
4(d)(i)	apply alternating p.d. (to / across crystal)	B1
	applying p.d. to / across crystal causes it to distort	B1
4(d)(ii)	$Z = \rho c$ $Z_m = 1100 \times 1600\ (= 1.76 \times 10^{6})$ $Z_b = 1900 \times 4100\ (= 7.79 \times 10^{6})$	C1
	intensity reflection co-efficient $= [(7.79 – 1.76) / (7.79 + 1.76)]^2$ $= 0.40$ or 40%	C1
	percentage transmitted = 60%	A1

Question	Answer	Marks